

INSULIN PROTEIN ANALYSIS USING UNIPROT AND BLASTP

Bioinformatics Project Report

Objective

To retrieve and study the human insulin protein from UniProtKB, obtain its FASTA sequence, perform a BLASTP similarity search against protein databases, and interpret sequence similarity, conserved regions and related proteins.

1. Protein Information

Protein Name	Insulin
UniProt Accession	P01308
Entry Name	INS_HUMAN
Organism	Homo sapiens (Human)
Gene Name	INS
Protein Length	110 amino acids
Protein Status	Reviewed (Swiss-Prot)
Protein Existence	Evidence at protein level
Function	Decreases blood glucose concentration and increases cellular uptake/permeability for monosaccharides

2. UniProt Search and Entry

A UniProtKB search for “insulin” returned multiple insulin-related proteins. The human insulin entry P01308 (INS_HUMAN) was selected for detailed analysis. The entry is reviewed in Swiss-Prot and contains 110 amino acids.

3. UniProt Search Results

The screenshot shows the UniProtKB search results for the query "insulin". The search bar at the top contains "insulin" and the results are displayed in a table. The table has columns for Entry, Entry Name, Protein Names, Gene Names, Organism, and Length. The first entry, P01308, is highlighted in green and corresponds to INS_HUMAN, Insulin, encoded by the INS gene in Homo sapiens (Human), with a length of 110 AA. Other entries include P06213 (INSR_HUMAN, Insulin receptor), P14735 (IDE_HUMAN, Insulin-degrading enzyme), P01317 (INS_BOVIN, Insulin), P01321 (INS_CANLF, Insulin), P01329 (INS_CAVPO, Insulin), and P17715 (INS_OCTDE, Insulin).

Entry	Entry Name	Protein Names	Gene Names	Organism	Length
P01308	INS_HUMAN	Insulin[...]	INS	Homo sapiens (Human)	110 AA
P06213	INSR_HUMAN	Insulin receptor[...]	INSR	Homo sapiens (Human)	1,382 AA
P14735	IDE_HUMAN	Insulin-degrading enzyme[...]	IDE	Homo sapiens (Human)	1,019 AA
P01317	INS_BOVIN	Insulin[...]	INS	Bos taurus (Bovine)	105 AA
P01321	INS_CANLF	Insulin[...]	INS	Canis lupus familiaris (Dog) (Canis familiaris)	110 AA
P01329	INS_CAVPO	Insulin[...]	INS	Cavia porcellus (Guinea pig)	110 AA
P17715	INS_OCTDE	Insulin[...]	INS	Octodon degus (Degu) (Sciurus degus)	109 AA

Figure 1. UniProtKB search results for "insulin", showing P01308 INS_HUMAN among the reviewed entries.

4. UniProt Protein Entry

The selected UniProt entry identifies insulin as a protein encoded by the INS gene in Homo sapiens. It provides curated functional annotation, sequence information and evidence for protein existence.

The screenshot shows the UniProtKB entry page for human insulin (P01308). The page is titled "P01308 · INS_HUMAN". It provides detailed information about the protein, including its function, names, taxonomy, subcellular location, disease & variants, PTM/processing, expression, interaction, structure, family & domains, sequence & isoform, similar proteins, and homologs. The function is described as: "Insulin decreases blood glucose concentration. It increases cell permeability to monosaccharides, amino acids and fatty acids. It accelerates glycolysis, the pentose phosphate cycle, and glycogen synthesis in liver." The gene ontology is also listed.

Field	Value
Protein	Insulin
Gene	INS
Status	UniProtKB reviewed (Swiss-Prot)
Organism	Homo sapiens (Human)
Amino acids	110 (go to sequence)
Protein existence	Evidence at protein level
Annotation score	5/5

Figure 2. UniProtKB entry page for human insulin (P01308).

5. Protein Sequence and FASTA

The canonical human insulin precursor sequence contains 110 amino acids. The FASTA sequence obtained from UniProt was used as the query for BLASTP.

The screenshot shows the UniProt entry for P01308 (INS_HUMAN). The left sidebar contains navigation links: Function, Names & Taxonomy, Subcellular Location, Disease & Variants, PTM/Processing, Expression, Interaction, Structure, Family & Domains, Sequence & Isoform, Similar Proteins, and Homologs. The main content area displays the 'Entry' tab, which includes the protein name, a description, and a list of tools (Download, Add, Highlight, Copy sequence). The canonical sequence is shown as a single line: MALWMRLRLPL LALLALWGPDPAAAFVNHQLCGSHLVEALYLVCGERGFFYTPKTRREAED LQVQVQLGGPGAGSLQPLALEGSLQKRGIVEQCCTSIICSLYQLENYCN. The sequence is annotated with residue numbers 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, and 110. The 'F8WCM5-1' isoform is highlighted, and a note states: 'The sequence of this isoform can be found in the external entry linked below. Isoforms of the same protein are often annotated in two different entries if their sequences differ significantly.'

Figure 3. UniProt sequence section showing the 110-amino-acid canonical sequence.

The screenshot shows the UniProt entry for P01308 (INS_HUMAN) in FASTA format. The sequence is displayed as a single line: >sp|P01308|INS_HUMAN Insulin OS=Homo sapiens OX=9606 GN=INS PE=1 SV=1 MALWMRLRLPL LALLALWGPDPAAAFVNHQLCGSHLVEALYLVCGERGFFYTPKTRREAED LQVQVQLGGPGAGSLQPLALEGSLQKRGIVEQCCTSIICSLYQLENYCN. The sequence is annotated with residue numbers 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, and 110.

Figure 4. FASTA format of human insulin P01308.

6. BLASTP Search

The FASTA sequence was submitted to NCBI BLASTP. BLASTP compares an amino-acid query sequence with protein sequences in a database and reports statistically significant similarities. The search shown here used the ClusteredNR database.

Figure 5. NCBI BLASTP input page with the human insulin FASTA sequence.

Figure 6. BLASTP search summary showing query length of 110 amino acids and the ClusteredNR database.

7. BLASTP Results and Significant Hits

The strongest matches are insulin proteins from closely related organisms. The first displayed hit, insulin preproprotein from Pan troglodytes, has 100% query coverage, 98% identity and an E-value of $1e-72$. Other primate insulin sequences also show high identity and complete coverage, demonstrating strong conservation.

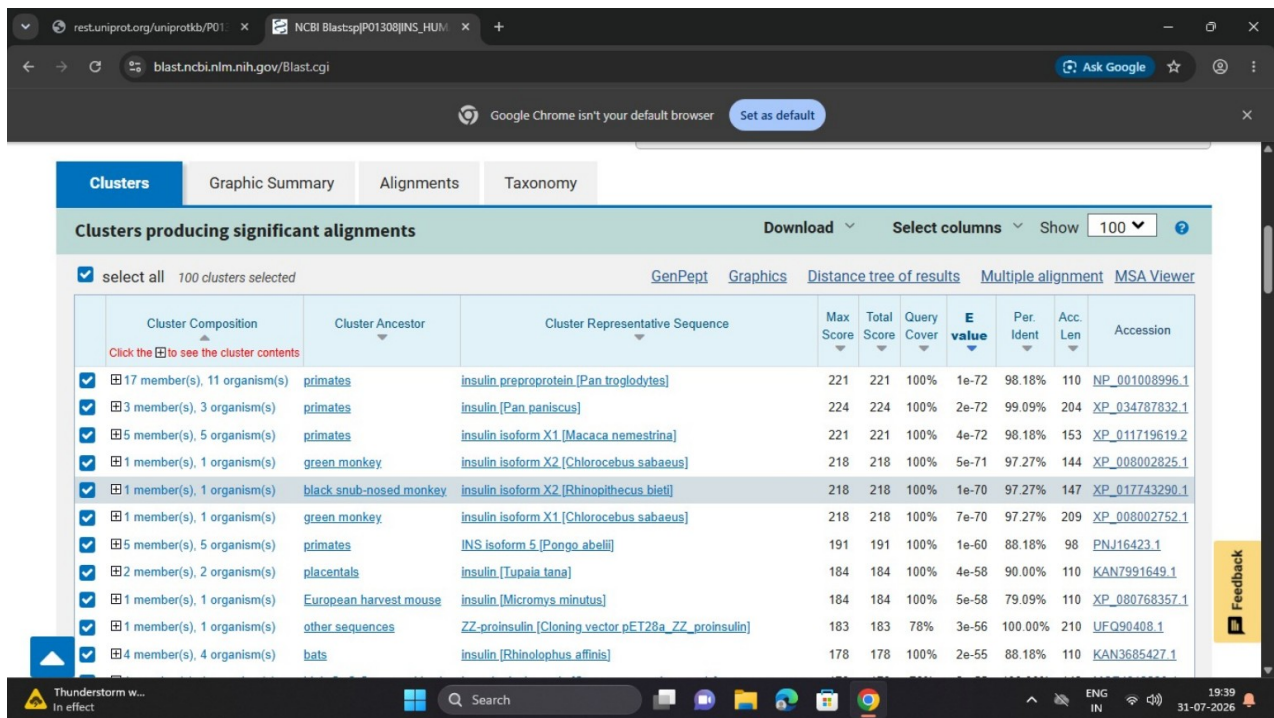


Figure 7. BLASTP cluster results showing significant insulin-related hits, query coverage, E-values and percentage identity.

8. Conserved Region and Graphic Summary

The graphic summary identifies a conserved IGF-like region across the query sequence. The distribution of BLAST hits shows that many sequences align across most or all of the 110-amino-acid query.

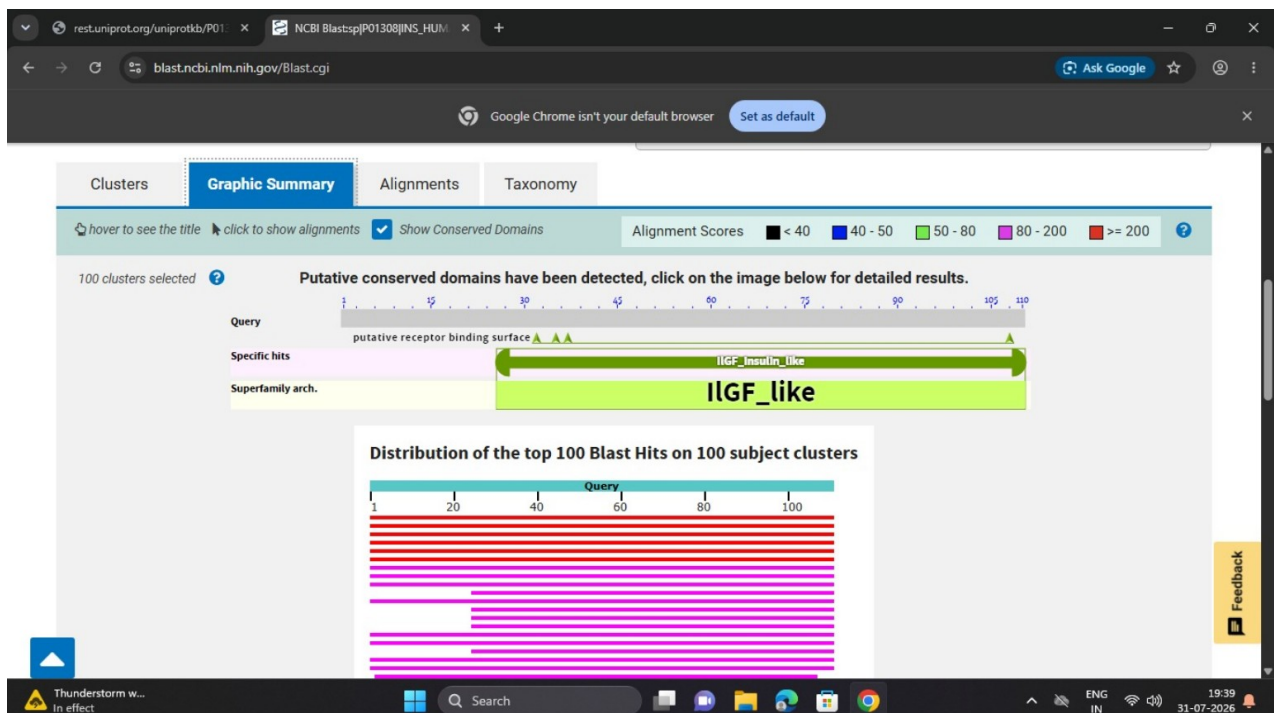


Figure 8. BLASTP Graphic Summary showing the conserved IGF-like region and distribution of top hits.

9. Pairwise Alignment

The pairwise alignment with Pan troglodytes insulin preproprotein shows 108 identical residues out of 110 (98% identity), 109/110 positives (99%), and 0 gaps. The complete query is aligned from residue 1 to 110,

indicating very high sequence conservation between human and chimpanzee insulin.

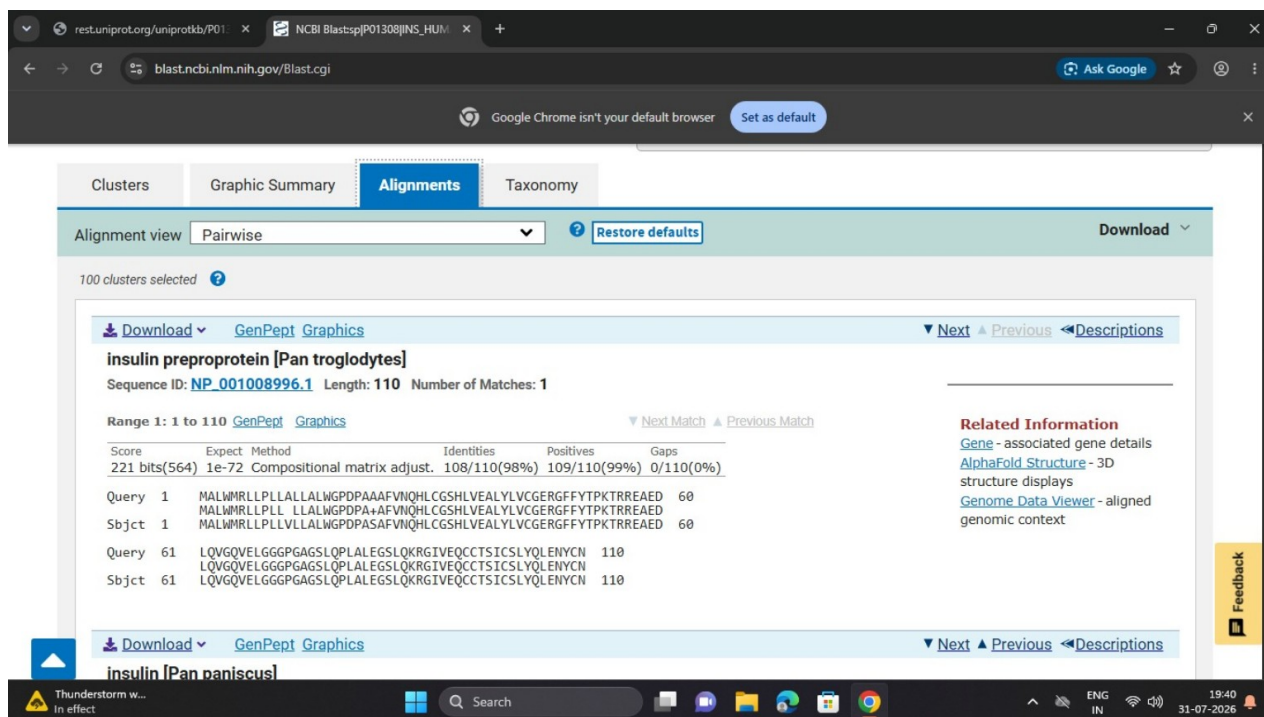


Figure 9. Pairwise BLASTP alignment of human insulin with Pan troglodytes insulin preproprotein.

10. Interpretation

Human insulin is a short and highly conserved protein. The strongest BLASTP matches occur among primates, reflecting their close evolutionary relationship. High percentage identity, complete query coverage and extremely low E-values indicate statistically significant and biologically meaningful similarity. The absence of gaps in the top pairwise alignment further supports strong conservation.

11. Conclusion

Human insulin (UniProt P01308) is a 110-amino-acid protein encoded by the INS gene. UniProt provides curated information about its function and sequence, while BLASTP demonstrates its similarity to insulin proteins from other species. Together, these tools are useful for protein identification, sequence comparison and evolutionary analysis.

12. Databases Used

UniProtKB/Swiss-Prot – human insulin, accession P01308. NCBI BLASTP – protein sequence similarity search. All screenshots in this report were supplied as part of the project workflow.